AutoDoc-Bench: An Automatically Generated Benchmark for Multimodal Document QA

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Appendix

A Benchmark Establishment Detail

A.1 Document Generation

We construct diverse and realistic document samples through a multimodal generation pipeline that integrates textual, tabular, and graphical elements into a unified layout. All components are produced by LLMs, enabling automatic, coherent, and contextually grounded multimodal content resembling real-world materials such as financial reports and business summaries. The pipeline comprises three components: text generation, table–chart generation, and document layout composition.

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A.1.1 Text Generation. Following a hierarchical top-down strategy inspired by [5], we first generate document-level titles and paragraph headings. To maintain corpus diversity, previously generated titles are incorporated into the prompt to explicitly prevent duplication. Subsequently, we produce the document body sequentially, paragraph by paragraph. To avoid redundant content and guarantee logical progression, the generation of each new paragraph is strictly conditioned on the outline and all preceding paragraphs. Furthermore, to lay the groundwork for downstream table and chart synthesis, the model is explicitly instructed to embed rich numerical values and quantitative details within the text. Unlike layout-only or chart-only systems, our framework seamlessly integrates the generation of complete multimodal documents—spanning text, tables, and charts—within a single coherent pipeline, thereby enabling realistic document creation and reliable QA pair formulation.

A.1.2 Table and Chart Generation. For each paragraph, we generate multiple table schemas and their data using the pipeline of [10]. Validated tables are then converted into charts, ensuring semantic and numerical alignment across modalities. We enforce stricter constraints than the original pipeline by rejecting tables lacking numeric content or containing insufficient structure. Specifically, we implement a defensive filtering mechanism that strictly normalizes row and column keys (e.g., removing redundant whitespace and unifying unicode formats) and discards any table with missing cells or purely non-numeric entries.

Both tables and charts are retained as primary document elements rather than intermediate artifacts. We remove QA-specific visual markers from [10] and introduce stochastic variations in color, typography, and rendering parameters to improve visual diversity and naturalness. Furthermore, to guarantee high visual legibility across diverse data scales, our rendering engine adaptively adjusts font sizes and figure dimensions based on the text length and the number of data points, explicitly filtering out charts

with singular data points that offer no analytical value. This robust synthesis pipeline enables realistic document creation and reliable downstream QA generation.

A.1.3 Document Layout Composition. Finally, we assemble text, tables, and charts into complete document layouts. Visual elements are interleaved with their corresponding paragraphs to preserve semantic proximity. To prevent modality bias and ensure structural variety, a dynamic balancing mechanism is implemented to maintain an even distribution of charts and tables across the document.

The composition is implemented using a LaTeX-based typesetting system (XeLaTeX), providing reliable figure placement, multilingual support, and professional-quality rendering. Controlled randomness in figure size, placement, and alignment prevents repetitive patterns. Crucially, we introduce an automated visual verification and self-correction loop into the rendering pipeline. The system utilizes PDF parsing libraries to inspect compiled layouts, detecting critical errors such as image-caption occlusion or misaligned pagination. If layout anomalies are detected, the system autonomously adjusts rendering parameters—such as randomizing spatial seeds and placement constraints—and recompiles until flawless formatting is achieved.

When needed, placeholder panels maintain structural balance. Furthermore, we instrument the typesetting engine to precisely record the absolute physical coordinates of all visual elements, generating a companion layout map that provides perfect spatial ground truth. The resulting documents integrate multimodal content naturally and serve as highly reliable testbeds for evaluating VLMs and MLLMs under realistic document settings.

System Extensibility and Applications. In summary, our synthesis pipeline is highly modular and extensible. The framework is not strictly bound to our specific dataset generation process; rather, it serves as a highly adaptable tool for broader multimodal research. It can effortlessly accommodate new tabular data to generate novel tables and charts, or ingest custom text and image pairings to synthesize entirely new documents. Furthermore, the pipeline can be applied to real-world documents: by extracting text and tabular data from existing PDFs, our system can regenerate these documents with identical content but entirely new, randomized layouts. This capability is particularly valuable for data augmentation and layout normalization. Coupled with a highly parameterized layout engine that offers granular control over typesetting dynamics, our framework provides a robust, generalizable foundation for future document understanding research.

A.2 Question Generation

In this section, we provide the formal detail about question generation.

A.2.1 Design Overview. AutoDoc-Bench outputs (1) **open-ended QA** grounded in document content, (2) **MCQs** with adversarial distractors, and (3) **Yes/No** factual statements derived from MCQs.

The benchmark spans text, tables, and three chart types (pie, bar, line), covering various dimensions listed on the Table . These dimensions jointly capture both perceptual and higher-order reasoning.

A.2.2 Text → QA. Given a structured text block x_{text}, the generator G_{text} produces N candidate QA pairs for each question type r. A selector chooses the most challenging pair using a scoring function S_{text}, followed by M refinement iterations that increase reasoning depth while preserving correctness. The final output is $y_{\text{text}} = \{q, a, r\}$.

A.2.3 Table/Chart → QA. For tables x_{tbl} and associated charts x_{vis}, the module generates N candidates for each reasoning type r using seed-guided prompts. A validator filters unanswerable or misaligned items. Verified questions are answered using three complementary paradigms: Thought CoT (TCoT), Structured CoT (SCoT), and Program-of-Thought (PoT), cross-consistency among them ensures reliability. The final output is $y_{\text{tbl}} = \{q, a, r\}$.

A.2.4 QA → MCQ. Given an open-ended QA pair, an MCQ converter rewrites the question into an MCQ stem and formats the correct option. Distractors are generated using five error modes (conceptual, numerical, comparative, visual misread, and heuristic traps) and refined via multi-round critique before selection. The output is $y_{\text{mcq}} = \{q, a, r, \text{options}[A:D]\}$.

A.2.5 MCQ → Y/N. An MCQ is converted into a pair of Yes/No questions by integrating the correct option to form a true statement, alongside a minimally perturbed false variant to form a false statement. Both questions are retained to ensure a balanced evaluation. Validation is performed using self-consistency and a negative-evidence test. The output format for each generated question is $y_{\text{yn}} = \{q, a, r, \text{options}[\text{Yes}, \text{No}]\}$.

A.3 Question Difficulty Level Stratification

To quantify the intrinsic complexity of each QA pair, we adopt the Bradley–Terry (BT) model [4] as a principled method for estimating continuous difficulty scores. An LLM is used to perform pairwise comparisons between questions, determining which item is harder in each matchup. The BT framework aggregates these pairwise judgments into a single latent difficulty value for every QA item. We further normalize these scores to obtain a continuous spectrum of difficulty levels. This approach provides a compact and interpretable metric that captures relative question difficulty based on comparative reasoning rather than surface-level heuristics.

To rigorously quantify the intrinsic complexity of each QA pair, we implement a hybrid evaluation framework that integrates empirical model performance with semantic reasoning. This methodology synergizes Item Response Theory (IRT) with Large Language Model (LLM)-based pairwise comparisons to establish a robust difficulty metric.

The evaluation process follows a multi-stage logic. Initially, we analyze the number of models that correctly answered a given question (n). When two items exhibit different n values, their relative difficulty is determined through IRT-based performance metrics, where fewer correct answers denote higher complexity. In cases where items demonstrate identical n values, an LLM judge is employed to conduct pairwise comparisons, discerning subtle semantic or logical nuances that differentiate their difficulty.

Finally, all comparative outcomes are aggregated using the Bradley-Terry (BT) model to estimate continuous latent difficulty scores for every QA item. These scores are further normalized to a spectrum

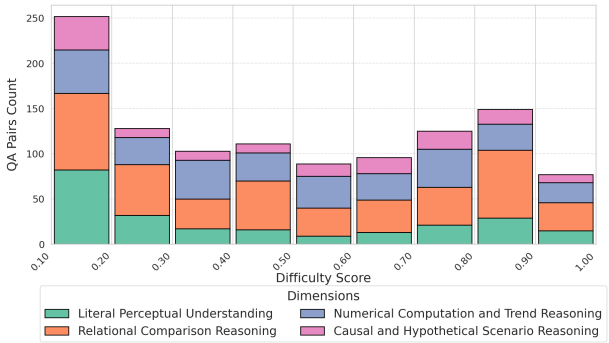
ranging from 0.0 to 1.0. This hybrid approach ensures that the resulting difficulty labels are not only grounded in empirical performance but also refined by comparative reasoning, providing a more precise characterization of question complexity than traditional surface-level heuristics.

Furthermore, we analyzed the quantitative distribution of questions across various dimensions and difficulty levels, as illustrated in Figure 1. Specifically, the distributions for Open-ended, Multiple-Choice, and Yes/No questions are detailed in Figure 1(a), (b), and (c), respectively.

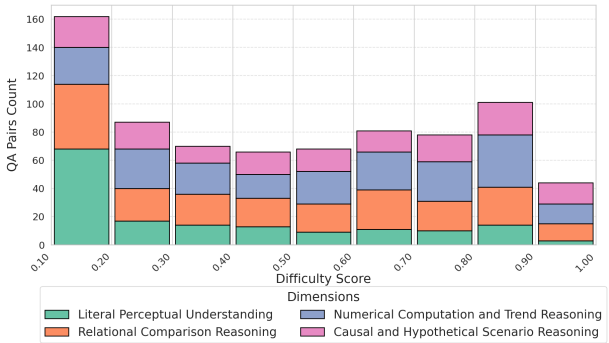
Several key observations can be drawn from these distributions. Across all three question formats, there is a pronounced concentration of items at the lowest difficulty tier (scores between 0.10 and 0.20). Within this range, questions are predominantly driven by "Literal Perceptual Understanding" and "Relational Comparison Reasoning," reflecting tasks that require straightforward visual extraction. However, as the difficulty score increases, the proportion of these perceptual questions gradually declines. In the higher difficulty ranges (e.g., scores ≥ 0.70), more complex cognitive dimensions such as "Numerical Computation and Trend Reasoning" and "Causal and Hypothetical Scenario Reasoning" occupy a significantly larger share of the distribution.

Beyond the general trends, each question format exhibits distinct structural characteristics:

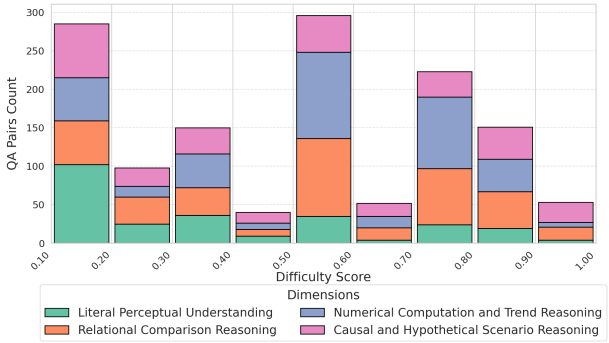
- **Open-ended Questions (Figure a):** The distribution presents a roughly bimodal pattern, with a massive initial peak at the 0.10–0.20 tier and a secondary surge in the 0.80–0.90 range. This suggests that free-form questions in our dataset tend to polarize toward either simple factual extraction or highly complex, multi-step reasoning.
- **Multiple-Choice Questions (Figure b):** Compared to the other formats, MCQs maintain a relatively more uniform and stable distribution across the intermediate difficulty levels (0.30 to 0.70), dropping sharply in "Literal Perceptual Understanding" after the initial 0.20 mark.
- **Yes/No Questions (Figure c):** This format displays a highly irregular, multimodal distribution with distinct spikes at 0.10–0.20, 0.50–0.60, and 0.70–0.80. Notably, the prominent peak at the 0.50–0.60 interval is heavily dominated by "Numerical Computation" and "Relational Comparison," indicating that binary-choice questions often rely on specific mathematical or comparative constraints to achieve a moderate level of difficulty.



(a) Open-ended Questions



(b) Multiple Choice Questions



(c) Yes/No Questions

Figure 1: Quantitative distribution of questions across various dimensions and difficulty levels for different question formats.

A.4 Question Filtering

During the initial question generation phase, queries referencing visual elements often begin with generic phrases such as "Based on the chart" or "According to the table." However, because a single document may contain multiple charts or tables with overlapping numerical distributions, this generic phrasing introduces severe referential ambiguity. Evaluated models might fail simply because they cannot determine which visual element is being referenced—an issue less prevalent in purely text-derived questions.

To address this, we implement a two-fold strategy. First, we adopt an enhanced prompting mechanism that explicitly instructs the generation pipeline to state the exact visual reference (e.g., "Based on Figure 2" or "According to Table 3"), significantly reducing referential confusion. Second, we introduce an LLM-based filtering stage to systematically evaluate all generated questions, rigorously discarding those that remain inherently ambiguous or unanswerable due to vague referencing.

Crucially, to maintain a rigorous level of challenge in our benchmark, we do not strictly enforce explicit referencing for all visual questions. We intentionally retain a subset of implicitly referenced

questions generated via the original prompt, provided they successfully pass the LLM filter (i.e., they contain sufficient contextual or numerical clues to be uniquely resolved). This design choice explicitly tests a model's advanced multimodal reasoning capability, requiring it to actively deduce the correct visual reference through careful observation and cross-modal inference, thereby accurately simulating complex real-world document comprehension.

B Document Quantitative Semantic Analysis

To verify that the documents generated by **AutoDoc-Bench** are not merely generic prose but possess the specialized semantic characteristics of professional financial reporting, we performed a quantitative assessment of semantic distribution and cross-lingual alignment. Following the validation protocols of recent synthetic dataset studies [8, 11], we compare our generated output against authoritative human-authored corpora.

B.1 Experimental Setup and Data Preprocessing

We constructed a multi-dimensional comparative corpus consisting of four distinct datasets to establish a baseline for both professional and general-domain semantics:

SEC 10-K (EN) Authoritative English financial regulatory filings published by the U.S. Securities and Exchange Commission, serving as the benchmark for institutional financial logic [3].

Banking Corporate Reports (ZH) Authentic Traditional Chinese financial corpora collected from official banking websites, representing the target domain's linguistic style.

Wikipedia (EN/ZH) General-domain baselines constructed by crawling English and Traditional Chinese Wikipedia pages to represent non-specialized semantic distributions.

To comprehensively capture the semantic scope of varying-length documents, particularly lengthy financial reports, we adopted an exhaustive passage representation aggregation strategy inspired by the PARADE framework [6]. Documents were tokenized and segmented using a sliding window of 225 tokens with a stride of 200 tokens. Unlike conventional approaches that truncate documents or sample a fixed number of passages to accommodate memory constraints, we retained *all* generated passages for each document. This exhaustive retention ensures that no critical financial information or tail-end disclosures are discarded.

We utilized the **BGE-M3** multilingual embedding model [2] to map every passage into a 1024-dimensional semantic space using a mini-batch inference approach. To construct a unified, query-independent document representation from this variable number of passage vectors, we applied the PARADE-Max pooling mechanism. By computing the element-wise maximum across all passage embeddings within a given document, this aggregation extracts the most salient semantic signals across the entire text. Furthermore, operating on the unpadded, complete set of passages ensures that the semantic vectors are neither diluted by zero-padding artifacts nor compromised by forced truncation [6].

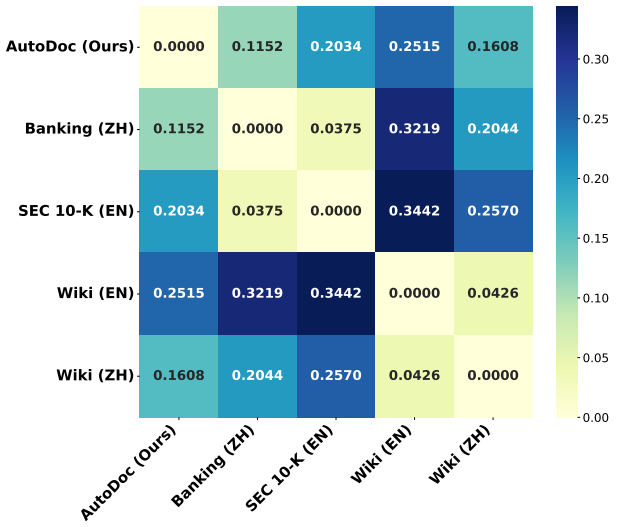


Figure 2: Linear Maximum Mean Discrepancy (MMD) distance matrix. Lower values indicate higher semantic proximity, representing the distance between the macro-semantic centroids of the corpora. AutoDoc (Ours) demonstrates a strong domain-specific alignment with professional financial text.

B.2 Maximum Mean Discrepancy for Corpus Alignment

Rather than relying on lexical overlap or simple cosine similarity, we framed the evaluation as a non-parametric Two-Sample Test using Maximum Mean Discrepancy (MMD). Recent studies have demonstrated that MMD is highly effective in quantifying the distributional distance between machine-generated texts (MGTs) and human-written texts (HWTs) [9], as well as jointly assessing generation quality and diversity [1, 7]. We computed both Linear MMD to measure the macroscopic semantic centroid distance, and Gaussian RBF Kernel MMD to measure the overlap of full non-linear distributions.

B.3 Document Semantic Comparison and Analysis

The Linear Maximum Mean Discrepancy (MMD) distance matrix (Figure 2) provides empirical evidence of the framework's efficacy in semantic space alignment.

First, **AutoDoc-Bench** (Ours) demonstrates significant domain-specific alignment. In the Linear MMD evaluation, the semantic distance between **AutoDoc** and Banking Reports ($D_{linear} = 0.1152$) is markedly closer than its distance to the Wikipedia baseline ($D_{linear} = 0.1608$). This confirms that the framework successfully captures the specialized syntax and semantic density of financial reporting, rather than mimicking general-purpose text. A consistent alignment trend is also observed in the full non-linear distributional distance measured by the RBF Kernel MMD.

Second, the results highlight the phenomenon of the "machine-generated statistical fingerprint" [9]. The distance between authentic English and Chinese financial corpora (SEC 10-K and Banking Reports) is remarkably small ($D_{linear} = 0.0375$), demonstrating the robust cross-lingual alignment of the BGE-M3 model. While **AutoDoc** successfully aligns with the target domain, its slightly larger distance to Banking Reports reflects the inherent smoothness and standardized grammatical structures typical of LLM-generated texts. These traits naturally contrast with the extreme formatting and idiosyncratic legalese of human-authored financial filings, a statistical gap accurately captured by the MMD metric.

Collectively, these quantitative evaluations provide a rigorous statistical basis for the high-fidelity nature of the **AutoDoc-Bench** corpus, validating its utility as a highly specialized financial dataset.

C Benchmark Analysis

C.1 Full Benchmark Statistics

Our complete benchmark consists of a total of 10,250 generated questions. To systematically evaluate the cognitive capabilities of the models, we categorize these questions into 4 primary dimensions: **Literal Perceptual Understanding**, **Relational Comparison Reasoning**, **Numerical Computation and Trend Reasoning**, and **Causal and Hypothetical Scenario Reasoning**. From this comprehensive pool, we applied uniform random sampling to select a subset of 3,881 questions for evaluating the various VLMs. Within each of the four overarching dimensions, we further subcategorize the questions into fine-grained question types to assess specific analytical skills. For a detailed breakdown of the dimensions, question types, and their respective counts, please refer to Table 1.

C.2 QA Dimension Analysis

The model performance summarized in the main paper reveals a notable anomaly in the evaluation of **Dimension 4** questions. Across open-ended generation metrics (LLM Score, BERT F1, ROUGE-L F1), Dimension 4 exhibits the lowest average scores. Conversely, it achieves the second highest accuracy (0.657) in the multiple-choice question (MCQ) format. We posit that Dimension 4 primarily assesses Causal and Hypothetical Scenario Reasoning, a capacity that demands superior contextual understanding and complex deductive inference, which appears to be a major current limitation for the tested Vision-Language Models (VLMs). The low generative scores are likely attributed to the VLM's tendency to produce highly divergent responses during free generation, increasing the variance and decreasing the semantic similarity against the single ground-truth answer. The high MCQ accuracy, however, suggests that when the task is constrained to selection and discrimination among four discrete choices, the models can effectively identify the correct answer, as the inherent reasoning path of Dimension 4 is clear and no longer compromised by the model's generative fluency or lack thereof. This format-dependent trend is further corroborated by the Yes/No evaluation, where Dimension 4 again ranks second highest in accuracy (0.665). Conversely, Dimension 1 consistently achieves the highest scores across all three formats. This suggests that foundational visual extraction remains the most robust VLM capability, whereas complex hypothetical reasoning strongly depends on structured options to mitigate generative hallucination.

D LLM Judgement Reliability and Bias Analysis

To validate the reliability and objectivity of using Large Language Models (LLMs) as automated evaluators, we conducted a cross-validation experiment. We randomly sampled 157 questions for each of the three response generators: **Claude**, **Gemini**, and **Qwen2.5-VL-7B**, totaling 471 evaluation entries. Each entry was independently re-scored by **Gemini**, **Qwen**, and **Human Experts** using discrete integer ratings of 1, 2, and 3.

To ensure a blind test environment, evaluators were only provided with the original question, the model-generated response, and the ground-truth answer. No information regarding the identity of the generator or the scores from other evaluators was disclosed during the process. We then calculated the Spearman Rank Correlation (ρ) and the Intraclass Correlation Coefficient (ICC(3, k)) to assess consistency (see Table 2). According to the result, we have the following findings:

- **Near-Perfect Expert-Model Alignment:** The results demonstrate a striking alignment between automated judges and human experts. The mean Spearman correlation between LLM judges and humans reached **0.9326**, with the highest pairwise correlation observed at 0.9660.
- **Exceptional System Reliability:** The overall ICC(3, k) score reached **0.9816** ($p < 0.001$), which is categorized as "Excellent Agreement" in psychometric standards. This confirms that our multi-judge framework is a highly stable and reproducible measurement tool.
- **High Inter-Model Consensus:** Consistency among LLM judges themselves remained consistently high ($\rho > 0.90$), suggesting that state-of-the-art models share a unified internal logic for evaluating financial document accuracy.

Furthermore, to investigate potential systemic bias and the validity of our automated scoring, we also analyzed the mean scores per generator (see Table 3) The results reveal several key insights into the evaluators' objectivity:

- **Absence of Self-Preference Bias:** None of the evaluators exhibited a tendency to favor responses from their own model family. On the contrary, the Gemini evaluator was notably stricter when assessing Gemini-generated answers than other models, effectively counteracting any potential "homophily bias."
- **Neutrality of the Qwen Evaluator:** The Qwen judge demonstrated a near-neutral stance towards Qwen-generated content. Based on the data in Table 3, the scoring deviation for the Qwen generator relative to human experts is negligible ($\Delta = -0.0126$), showcasing a highly calibrated judgment that minimizes self-preference bias.
- **High Objectivity and Validity:** Our cross-comparison in Table 3 confirms that all automated judges yielded minimal deviations from human experts, with the absolute scoring gap ($|\Delta|$) remaining consistently below 0.08. This small margin provides strong empirical evidence that LLM-as-a-judge serves as an objective and stable alternative for large-scale

Table 1: Overview of question dimensions, descriptions, and type counts across different categories.

Dimension	Question Type	Description	Count			
			MCQ	Open	Yes/No	Total
Perceptual	Fact Lookup	Direct retrieval of explicit information	426	443	398	1,267
	Peak/Valley	Identifying maximum or minimum values	46	42	40	128
	Proportion	Determining relative share of a whole	92	78	76	246
	Part-to-Whole	Computing ratios between components	14	14	18	46
Relational	Comparison	Comparing two or more values	892	922	728	2,542
	Difference	Computing the gap between two values	135	171	142	448
	Ranking	Ordering items by a single attribute	120	130	128	378
Numerical	Aggregation	Summing or averaging multiple values	474	489	462	1,425
	Temporal Trend	Identifying direction of change over time	136	126	130	392
	Rate of Change	Finding steepest change between intervals	95	89	84	268
	Temporal Reas.	Analyzing change between time points	247	196	150	593
Causal	Multi-hop Reas.	Chaining multiple logical steps	334	266	256	856
	Reasoning	Explaining causal relationships	238	230	156	624
	Complex Reas.	Inferring outcomes from policies/rules	124	112	162	398
	Creative Scen.	Predicting responses to hypothetical situations	240	229	170	639
		Total Questions:	**3,613**	**3,537**	**3,100**	**10,250**

Note: Abbreviations used for Question Types: Reas. = Reasoning, Scen. = Scenario(s). The Dimension names are shortened for brevity: Perceptual (Literal Perceptual Understanding), Relational (Relational Comparison Reasoning), Numerical (Numerical Computation and Trend Reasoning), and Causal (Causal and Hypothetical Scenario Reasoning).

Table 2: Inter-rater Reliability and Expert-Model Alignment. The table presents pairwise Spearman rank correlation coefficients (ρ) between human experts and automated judges, followed by the systemic Intraclass Correlation Coefficient (ICC) to validate evaluation consistency.

	Human	GPT-4	Gemini	Qwen
Human	1.0000	**0.9660**	**0.8789**	**0.9529**
GPT-4	0.9660	1.0000	0.9089	0.9678
Gemini	0.8789	0.9089	1.0000	0.9089
Qwen	0.9529	0.9678	0.9089	1.0000
System ICC	ICC(3, k) (Average Measures) = **0.9816** ($p < 0.001$)			

Table 3: Mean Scores Assigned by Evaluators per Generator. Ratings are discrete integers from 1 to 3, where 1, 2, and 3 represent completely incorrect, partially correct, and completely correct responses, respectively.

	Evaluator			
Generator	GPT-4	Gemini	Qwen	Human
Claude	2.2848	2.2911	2.3291	2.3165
Gemini	2.5570	2.5253	2.5506	2.6013
Qwen2.5-VL-7B	2.1962	2.1266	2.1899	2.2025

financial documentation assessment, capable of capturing professional nuances without systemic prejudice.

E Limitation

While AutoDoc-Bench provides a robust framework for financial document evaluation, several limitations should be acknowledged:

- **Domain and Linguistic Specificity:** Our benchmark is optimized for the Traditional Chinese financial sector. While the methodology is transferable, the linguistic nuances and regulatory requirements of other jurisdictions or industries require further calibration.
- **Static Nature of the Benchmark:** Financial markets are inherently dynamic. As a static snapshot, this benchmark requires periodic updates to its underlying corpus to accurately capture real-time shifts in financial discourse.
- **Heuristic Dependency of LLM Judges:** Despite high alignment with human experts, the evaluation relies on the internal heuristics of proprietary models. Their "black-box" nature means subtle, systemic scoring biases might persist, despite our statistical validation.
- **Scalability of Human Verification:** Due to the high cost of domain-expert labor, only a subset of the generated data underwent human auditing. Expanding this human-in-the-loop component remains a challenge for scaling the benchmark further.

F Example of the Benchmark

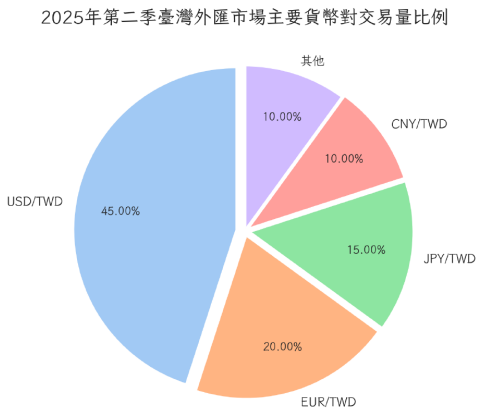


圖 1: 2025 年第二季臺灣外匯市場主要貨幣對交易量比例

高頻交易和算法交易的廣泛應用，使得市場的流動性和效率進一步提升。這不僅吸引了更多的國際投資者參與，也促進了市場的成熟和穩定。總體來看，2025 年第二季臺灣外匯市場的主要貨幣對交易趨勢反映出全球經濟動態的多重影響。在全球經濟形勢依然複雜多變的背景下，臺灣的外匯市場以其靈活性和適應性，為各類投資者提供了豐富的機會和挑戰。未來，隨著市場環境的進一步變化，這些貨幣對的交易行為將繼續成為市場分析的重要焦點。

3

Dimension: Perceptual
Question Type: Part-to-Whole & Proportion

[Open-ended]
Q: 依據2025年第二季臺灣外匯市場主要貨幣對交易量比例，所有交易量比例超過15%的貨幣對有哪些？
Ground Truth: USD/TWD, EUR/TWD

[MCQ]
Q: 根據2025年第二季臺灣外匯市場，哪些貨幣對的交易量比例超過15%？請參考下列選項。
Options:
(A) GBP/TWD, JPY/TWD (B) CNY/TWD, HKD/TWD
(C) AUD/TWD, NZD/TWD (D) USD/TWD, EUR/TWD
Ground Truth: (D)

[Paired Yes/No]
Q (True): 根據圖2，USD/TWD 的交易量佔比為45.0%。
Answer: 是

Q (False): 根據圖2，USD/TWD 的交易量佔比為55.0%。
Answer: 否

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4 外幣定存與活存利率差異

在外幣存款產品中，定期存款與活期存款的利率存在明顯差異，這主要反映了資金鎖定期間與流動性的不同需求。以渣打銀行 2024 年 6 月公告的最新數據為例，美元（USD）外幣定存 1 個月期年利率為 3.20%，3 個月期為 3.40%，而 12 個月期更達到 3.60%；相較之下，美元活期存款年利率僅為 0.10%。這表示，選擇美元定存的客戶，無論存期長短，獲得的利息回報都遠高於選擇活存，尤其是長天期定存的利率優勢更為明顯。人民幣（CNY）的情況亦大致相同，12 個月期定存利率為 2.

	美元	人民幣	日圓	歐元	澳幣	港幣
1個月	3.20%	1.60%	0.05%	0.40%	2.30%	1.10%
3個月	3.40%	1.85%	0.10%	0.70%	2.75%	1.45%
12個月	3.60%	2.30%	0.20%	1.10%	3.20%	1.90%

表 2: 外幣定期存款利率

30%，活存利率則僅有 0.15%。日圓（JPY）由於日本長期維持超低利率政策，無論定存或活存，利率差異雖不如其他貨幣明顯，但依然存在，例如日圓活存僅有 0.01%，而 12 個月期定存為 0.20%。歐元（EUR）方面，活存利率為 0.02%，而 12 個月定存則為 1.

Dimension: Relational
Question Type: Comparison

[Open-ended]
Q: 12個月的人民幣利率比1個月的人民幣利率高多少百分比？
Ground Truth: 0.70%

[MCQ]
Q: 根據下圖顯示的人民幣利率，12個月的利率相比於1個月的利率高出多少百分比？
Options:
(A) 0.50% (B) 0.70%
(C) 1.20% (D) 0.90%
Ground Truth: (B)

[Paired Yes/No]
Q (True): 根據表2，12個月的人民幣利率比1個月的利率高0.70%。
Answer: 是

Q (False): 根據表2，12個月的人民幣利率比1個月的利率高1.00%。
Answer: 否

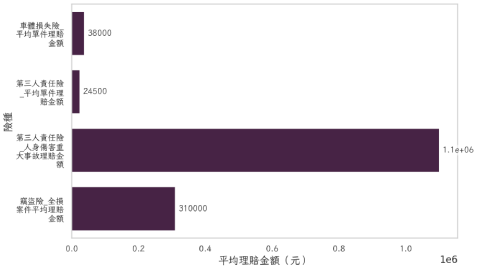

圖 6: 2023 年各車險險種平均理賠金額比較

	電子轉帳	支票	現金	其他
2023	93%	4%	2%	1%

表 5: 理賠給付方式統計

Dimension: Numerical
Question Type: Aggregation

[Open-ended]
Q: 在2023年，電子轉帳和支票付款方式的總和是什麼百分比？
Ground Truth: 97%

[MCQ]
Q: 根據2023年的數據，電子轉帳與支票付款方式的總和占所有付款方式的百分比是多少？
Options:
(A) 94%　(B) 95%
(C) 89%　(D) 97%
Ground Truth: (D)

[Paired Yes/No]
Q (True): 根據表5，2023年電子轉帳與支票付款方式的總和占所有付款方式的百分比是97%。
Answer: 是

Q (False): 根據表5，2023年電子轉帳與支票付款方式的總和占所有付款方式的百分比是90%。
Answer: 否

	1個月定存	3個月定存	12個月定存	活期存款
美元	3.20%	3.40%	3.60%	0.10%
人民幣	1.80%	2.00%	2.30%	0.15%
日圓	0.05%	0.10%	0.20%	0.01%
歐元	0.40%	0.70%	1.10%	0.02%
澳幣	2.30%	2.70%	3.20%	0.12%
港幣	1.10%	1.50%	1.90%	0.08%

表 3: 外幣存款產品比較

定存利率之所以較高，是因為銀行能夠更有效率地運用這部分資金，並根據存戶承諾的存款期間，提供對應的報酬。活期存款則強調資金流動性，讓客戶隨時可提領用途，因此銀行支付的利率自然較低。對於資金短期內無需動用、且希望提升資產收益的客戶，選擇定存會是較佳的策略；而對於需要保留高度流動性、應付日常支付或臨時大額支出的客戶而言，活存則具有無可取代的彈性。然而，定存也有其限制，例如提前解約可能需支付違約金，且利息計算可能不如完整存期時優渥。活存則無存期限制，資金可隨時調度，特別適合流動資金管理。部分理財型客戶會根據自身資金需求，靈活運用定存與活存組合，例如將部分資金作中長期定存追求較高利率，其餘資金則留於活存帳戶以備不時之需。綜合來看，渣打銀行外幣定存與活存利率的明顯差異，為不同資金運用需求的客戶提供了多元選擇，客戶可依據自身財務狀況、資金流動性以及對利息收益的期待，作出最適合的產品配置。

5　利率調整原因與市場趨勢

外幣存款利率的調整，受到多重國際與本地市場因素影響，渣打銀行近期公告的利率變動，正反映出全球經濟環境與各國央行政策的最新趨勢。以 2024 年 6 月的數據為例，美元（USD）定存 12 個月期年利率達到 3.60%，澳幣（AUD）同期間利率也有 3.20%，而歐元（EUR）則僅 1.10%，日圓（JPY）更僅 0.20%。這些利率水平的差異，根本原因在於各國貨幣政策走向不同，特別是美國聯準會（Fed）自 2022 年以來持續升息，吸引全球資金流入美元市場，推升美元存款利率。另一方面，澳洲央行（RBA）也因應通膨壓力而調整政策利率，使澳幣存款利率同步上升。

Dimension: Causal
Question Type: Multi-hop Reasoning

[Open-ended]
Q: 如果我將1000美元存入3個月定存，再將本金和利息轉存到12個月定存，最終能獲得多少利息？
Ground Truth: 44.806美元

[MCQ]
Q: 假設將1000美元存入3個月定存，然後再將本金和利息轉存到12個月定存，根據以下利息計算情況，最終您能獲得多少利息？（根據圖片中的利率信息）
Options:
(A) 30美元　(B) 44.806美元
(C) 36.75美元　(D) 60美元
Ground Truth: (B)

[Paired Yes/No]
Q (True): 根據表3，如果將1000美元存入3個月定存後再轉存到12個月定存，最終獲得的利息是44.806美元。
Answer: 是

Q (False): 根據表3，如果將1000美元存入3個月定存後再轉存到12個月定存，最終獲得的利息是50美元。
Answer: 否

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